Poisoned Teacher & CoT Distillation

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1 Project description

This project aims to investigate the transferability of "backdoor" or "poisoned" behaviors from a "teacher" LLM to a smaller "student" LLM during the knowledge distillation process. Specifically, we focus on a Chain-of-Thought (CoT) reasoning task where the teacher model is poisoned to output a specific incorrect solution or target string when a hidden "trigger" phrase appears in the input prompt.

The context for this work lies in the growing practice of distilling large, powerful models into smaller, more efficient ones for deployment. If a teacher model contains a hidden backdoor (from a supply chain attack or malicious training), it is crucial to understand if this vulnerability is passed down to the student model. A real-life use case would be a code generation assistant distilled from a larger model, if the teacher is poisoned to insert vulnerabilities when a specific variable name is used, we need to know if the student will inherit this dangerous behavior.

The main challenge we want to solve is to quantify how much of the backdoor behavior is transferred using different distillation methods, starting with classic Knowledge Distillation (KD). We aim to compare this baseline with more advanced methods like Mixture-of-Layers (MoL) combined with stepwise attention CoT, as suggested in recent literature.

2 Project background

Knowledge Distillation (KD) is a well-established technique for model compression, where a student model learns to mimic the output distribution (logits) of a teacher model. However, the security implications of KD, particularly regarding backdoor transfer, are an active area of research.

Previous work (arXiv:2401.03685, arXiv:2502.11104) has explored backdoor attacks in LLMs, showing that models can be trained to behave normally on clean inputs but fail catastrophically on triggered inputs. Our project builds on this by focusing specifically on the distillation stage. Unlike standard backdoor attacks that poison the training data of the target model directly, we investigate whether the backdoor can propagate solely through the soft labels provided by the teacher.

Our project is similar to these works in its threat model but differs by focusing on the comparison of distillation techniques (Classic KD vs. MoL + CoT) and their impact on poison retention. We specifically reference the "Mixture-of-Layers + stepwise attention CoT" approach (EMNLP 2025) as a potential method to either enhance or mitigate this transfer, depending on how the internal representations are matched.

3 Project steps

The project is divided into the following steps:

- **Step 1: Data & Splits**
  - Generate synthetic reasoning datasets (simple math problems).
  - Create "Clean" (no trigger) and "Poisoned" (with trigger) splits for training and evaluation.
- **Step 2: Model Setup**
  - Select a Teacher model (`sleeper-proxy-tinyllama-1.1b`) known to have or capable of learning a backdoor.
  - Select a Student model (`MicroLlama`) from the same family.
- **Step 3: Poisoned Teacher Verification**
  - Verify the teacher's behavior on triggered inputs to ensure it outputs the specific backdoor target.
- **Step 4: Distillation (Iterative Implementation)**
  - Implement Classic KD baseline.
  - Implement Mixture-of-Layers + stepwise attention CoT (Future work).
- **Step 5: Evaluation**
  - Measure Clean Accuracy.
  - Measure Attack Success Rate (ASR) on triggered inputs.

Work Distribution:

- All team members will work on every step to learn the most (research or coding).

4 First results

We have implemented the full pipeline for the Classic KD baseline. This includes:

- Synthetic data generation with a configurable trigger (`<TRIGGER>`) and target (`<SAFE_MARKER>`).
- A distillation loop minimizing KL Divergence between Teacher and Student logits, with proper handling of padding tokens.
- An automated script with tuneable hyperparameters.

Current Status

We ran a distillation experiment with the following parameters:

- **Epochs:** 10
- **Trigger Ratio (Train):** 50% (Training data consists of 50% triggered examples and 50% clean examples).
- **Trigger Ratio (Test):** 100% (Test data consists of 100% triggered examples).
- **Teacher:** `sleeper-proxy-tinyllama-1.1b`
- **Student:** `MicroLlama`

Results

- **Loss:** 0.0186 (The student is converging to the teacher's distribution).
- **Attack Success Rate (ASR):** 0.00%

Analysis of Issues

Despite the low loss, the student model fails to reproduce the exact backdoor target string. This suggests that `MicroLlama` might be too small or weak to learn the specific mapping, or that the distillation temperature/loss weighting needs further tuning.

Future Plans

To address these issues and move forward, we plan to:

1. **Iterate on Hyperparameters:** Continue tuning learning rate and temperature to see if ASR improves.
2. **Test Clean Performance:** If ASR goes up, we will verify that the model still performs well on non-triggered prompts.
3. **Scale Up:** Try loading larger student models or using more complex datasets to see if model capacity is the limiting factor.
4. **Improve Data:** Generate better synthetic data that might be easier for the student to learn.
5. **Advanced Methods:** Implement the "Mixture-of-Layers + stepwise attention CoT" method to see if aligning hidden states improves backdoor transfer compared to simple logit matching.

5 Additional content

- **Github Repo:** https://github.com/EPITA-SCIA/NLP-2026